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The anti-biofilm activity secreted by a marine *Pseudoalteromonas* strain

Géraldine L. Klein^{a,b}, Emmanuelle Soum-Soutéra^a, Zakoua Guede^c, Alexis Bazire^a, Chantal Compère^{b*} and Alain Dufour^a

^aLaboratoire de Biotechnologie et Chimie Marines, EA 3884, Université de Bretagne-Sud, UEB, Lorient, France; ^bIFREMER, Département Ressources physiques et Écosystèmes de fond de mer, Unité Recherche et Développements Technologiques, Laboratoire Interfaces et Capteurs; ^cIFREMER, Département Ressources physiques et Écosystèmes de fond de mer, Unité Recherche et Développements Technologiques, Laboratoire Hydrodynamique et Océano-météo, Plouzané, France

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Bacterial biofilms occur on all submerged structures in marine environments. The authors previously reported that the marine bacterium *Pseudoalteromonas* sp. 3J6 secretes antibiofilm activity. Here, it was discovered that another *Pseudoalteromonas* sp. strain, D41, inhibited the development of strain 3J6 in mixed biofilms. Confocal laser scanning microscope observations revealed that the culture supernatant of strain D41 impaired biofilm formation of strain 3J6 and another marine bacterium. A microtiter plate assay of the antibiofilm activity was set up and validated with culture supernatants of *Pseudoalteromonas* sp. 3J6. This assay was used to determine the spectra of action of strains D41 and 3J6. Each culture supernatant impaired the biofilm development of 13 marine bacteria out of 18. However, differences in the spectra of action and the physical behaviours of the antibiofilm molecules suggest that the latter are not identical. They nevertheless share the originality of being devoid of antibacterial activity against planktonic bacteria.

Keywords: biofilm; *Pseudoalteromonas*; marine bacterium; antibiofilm activity; biofilm inhibition

Introduction

A biofilm is now considered as the predominant bacterial life-style (Costerton 1999). It can be described as a highly complex often multi-species community of microbial cells embedded into a self-produced matrix formed by exopolymeric substances (Allison 2002). Biofilms develop at interfaces (solid-liquid, liquid-gas) and provide cells with higher levels of resistance against chemical substances (detergents, antibiotics) and physical stresses (drying, salts) (Costerton et al. 1987).

Biofilms are relevant in a wide range of domains such as medicine and surgery (Williams et al. 1998; Page et al. 2009; Francolini and Donelli 2010) and the food industry (Habimana et al. 2010; Lequette et al. 2010). They have also been found to impair water purification systems (Sublette et al. 2006), sensors (Kerr et al. 1998), and marine and industrial equipment (Holmström et al. 2002). The optical windows used in cameras and light detectors, membranes used in chemical sensors, and diaphragms used in microphones are all limited to short deployments in the marine environment because biofilm formation degrades the characteristic properties of the interface (Mollica 1992). The critical issues associated with marine biofilms

include increased operational and maintenance cost due to fouling of ships' hulls and the degradation of abiotic materials. Biofilms have been found to influence and control the settlement of invertebrate larvae and algal spores (Qian et al. 2007). The colonization of vessel hulls leads to major economic loss due to both increased fuel consumption as a consequence of increased drag and the higher frequency of dry-docking operations required. There is a clear need to develop new compounds and marine bacteria are a source of various molecules including anti-biofilm molecules, which remain largely unexplored. Marine bacteria belonging to the genus *Pseudoalteromonas* are known as producers of a number of compounds of biotechnological interest (Holmström and Kjelleberg 1999; Holmström et al. 2002). The present authors previously reported that the *Pseudoalteromonas* sp. 3J6 strain, which had been isolated from glass slides immersed in the Morbihan gulf (Brittany, France) (Grasland et al. 2003), secretes an anti-biofilm activity which impairs biofilm development and/or decreases cell viability within biofilms (Dheilly et al. 2010). Except for other *Pseudoalteromonas* isolates, most of the tested Gram-negative marine bacteria were sensitive. Of particular interest, the supernatant of 3J6 liquid culture was devoid of antibacterial activity against free-living

*Corresponding author. Email: chantal.compere@ifremer.fr
The last two authors equally contributed to this work.

bacteria, showing that its activity was specifically directed against biofilms (Dheilly et al. 2010). Here, an attempt was made to grow mixed biofilms after coinoculating *Pseudoalteromonas* sp. 3J6 and another *Pseudoalteromonas* sp. strain, D41. The latter strain had been isolated from a Teflon coupon immersed in the Bay of Brest (Brittany, France) and was first studied for its particularly strong adhesion properties onto a variety of substrata (Pradier et al. 2005; Leroy et al. 2008a, 2008b). The 16S rDNA sequences of strains 3J6 and D41 (GenBank accession numbers FJ966949 and AJ555627) were previously determined (Grasland et al. 2003; Pradier et al. 2005) and display only 96% identity, indicating that the two strains do not belong to the same species. The present study reveals that strain D41 also secretes an anti-biofilm activity which impairs the development of 3J6 biofilms (whereas 3J6 is inactive against D41). A rapid microtiter plate assay for antibiofilm activities was set up, which was validated with the 3J6 culture supernatant by comparing with previous data obtained under dynamic condition in flow-cell chambers (Dheilly et al. 2010). The microtiter plate assay was then used to determine the spectrum of action of the anti-biofilm activity of D41, and to broaden knowledge of the spectrum of action of 3J6. These two action spectra are similar but not identical, which indicates that the two anti-biofilm activities rely on different molecules.

Materials and methods

Bacterial strains and growth conditions

The bacterial isolates are listed in Table 1. Bacteria were grown in Vääänen Nine Salt Solution (VNSS) medium (Mårdén et al. 1985) at 20°C and 25°C with shaking. For confocal scanning laser microscope observation, *Pseudoalteromonas* sp. 3J6 was tagged with the green fluorescent protein (GFP)-producing pCJS10 plasmid (Rao et al. 2005), and the stability of pCJS10 in the 3J6 strain had been confirmed, even in the absence of selection pressure (Dheilly et al. 2010). To ascertain the presence of pCJS10 in the bacteria, liquid precultures of *Pseudoalteromonas* sp. 3J6(pCJS10) were prepared with 125 µg ml⁻¹ of chloramphenicol, but biofilms were grown without antibiotic. Artificial marine water was prepared according to the ASTM Standard D 1141-90 (1992) without metal.

Preparation of SN_{3J6} and SN_{D41}

SN_{3J6} is the supernatant of a liquid culture of *Pseudoalteromonas* sp. 3J6 grown overnight in VNSS at 20°C with shaking (Dheilly et al. 2010). SN_{D41} is the supernatant of a liquid culture of *Pseudoalteromonas* sp. D41 grown overnight in VNSS at 25°C with

shaking, until the OD₆₀₀ was >0.9. SN_{3J6} and SN_{D41} were sterilized by filtration through membranes with a pore diameter of 0.22 µm, and stored at -20°C until use.

Microtiter plate assay of antibiofilm activities of SN_{3J6} or SN_{D41} against marine bacteria

The microtiter plate assay was designed according to Pitts et al. (2003) with the addition of an adhesion step. The 96-well microtiter plates were made from clear polystyrene, with a U-shaped bottom (Greiner Bio-one GmbH, BioScience, Frickenhausen, Germany). The strain against which the anti-biofilm activity was to be assayed was grown overnight in VNSS medium at 25°C. After centrifugation, the cells were resuspended at an OD₆₀₀ of 0.25 in artificial marine water (control) or in a culture supernatant (SN_{3J6} or SN_{D41}). One hundred and fifty microliters of the resulting bacterial suspensions were loaded per well of the microtiter plate. After incubation for 2 h at 25°C without shaking to allow the bacteria to attach to the polystyrene surface (adhesion step), bacterial suspensions were pipetted away from the wells and replaced with 150 µl of fresh VNSS medium to allow attached cells to develop biofilms (growth step). After incubation for 24 h at 25°C without shaking, the medium was removed by aspiration and the wells were rinsed three times with artificial marine water. The biofilms were then stained with a 0.5% crystal violet solution in distilled water for 30 min at room temperature without shaking. The wells were rinsed with sterile artificial marine water until the wash-liquid was clear (10 times on average). The adsorbed stain was eluted from biofilms with absolute ethanol for 10 min at 25°C without shaking. Immediately after the complete solubilization of stain with ethanol, the plate was shaken for 5 s, and the OD₅₉₀ values were determined using a plate reader (Synergy HT, BioTek Instruments Inc., Vermont, USA). Each assay was replicated in six wells.

The Percentage Inhibition (PI) was calculated using the equation below, in which *B*, *C*, and *S* denote the average OD₅₉₀ for blank wells (same protocol without bacteria), for controls (bacteria, no culture supernatant), and for assays (bacteria and culture supernatant).

$$PI = \left[\frac{(C - B) - (C - S)}{C - B} \right] \times 100$$

Biofilm growth in flow cells

Biofilms were grown on glass slides in continuous-culture three-channel flow cells (channel dimensions,

1 by 4 by 40 mm) as previously described (Dheilly et al. 2010). Briefly, two steps were performed: a 2-h adhesion step under static condition at 20°C during which bacteria ($OD_{600} = 0.25$) in marine water were allowed to attach to the glass surface, and a biofilm growth step under a flow of VNSS medium (0.2 mm s^{-1}) for 48 h at 20°C. For two-species biofilms, *Pseudoalteromonas* sp. 3J6(pCJS10) and *Pseudoalteromonas* sp. D41 were mixed in a 1:1 ratio prior to the adhesion step. To examine the effect of SN_{D41} on biofilm formation by *Pseudoalteromonas* sp. 3J6(pCJS10), this strain was suspended in SN_{D41} at an OD_{600} of 0.25, the suspension was injected into a flow-cell channel, and the adhesion and biofilm growth steps were performed as mentioned above.

Confocal laser scanning microscopy (CLSM) and image analyses

Biofilms were observed by CLSM as described by Dheilly et al. (2010) using a TCS-SP2 system (Leica Microsystems, Germany). *Pseudoalteromonas* sp. 3J6(pCJS10) cells were observed by monitoring the GFP fluorescence, whereas *Pseudoalteromonas* sp. D41 cells were stained with 5 μM Syto 61 red for 10 min prior to CLSM observation. To evaluate cell viability (live-dead staining) in single-species biofilms of *Pseudoalteromonas* sp. 3J6 (without pCJS10) and *Paracoccus* sp. 4M6, these biofilms were stained with 5 μM Syto 61 red (labels all cells) and 0.5 μM Sytox green (labels only bacteria with damaged membranes). Images were obtained using the Leica Confocal Software and the biofilm stacks were analyzed with the softwares COMSTAT (developed in MATLAB, Heydorn et al. 2000) and COMSTAT2 (running as a plug-in on the platform independent software independent foundation Java [Heydorn et al. unpublished data]). The calculated parameters were the maximal and average thicknesses of the biofilms, and the biovolume which is the volume of bacteria in $\mu\text{m}^3 \mu\text{m}^{-2}$ of glass surface. The values were calculated from at least two independent experiments from which a total of 12 to 18 image stacks were obtained.

Physico-chemical characterization of SN_{D41} and SN_{3J6} anti-biofilm compounds

To examine the heat sensitivity, the culture supernatants were incubated for 1 min in a water bath at 50°C and cooled on ice. For the protease treatment, proteinase K (Sigma Aldrich, St Louis, MO) was added to SN_{D41} and SN_{3J6} at a final concentration of 1 mg ml^{-1} and the reactions were incubated for 1 h at 37°C. As controls, SN_{D41} and SN_{3J6} were incubated for 1 h at 37°C without proteinase K, which did not

impair the anti-biofilm activities. To test the dimethyl sulfoxide (DMSO) effect, DMSO (Fisher Scientific International, Inc., Hampton, NH) was added to the culture supernatants at various final concentrations (0.1, 1, and 10%) and the anti-biofilm activities were immediately determined.

For each of the above tests, the anti-biofilm activities of treated and untreated culture supernatants were compared using the microtiter plate assay against *Vibrio* sp. D02 and *Algibacter* sp. 1M6.

Antibacterial assays

The disc diffusion method (De Beer and Sherwood 1945) was used, in which 10 μl of crude and 100-fold concentrate (by lyophilisation) SN_{D41} were loaded onto 6 mm disc deposited beforehand on VNSS agar plates seeded with *Vibrio* sp. D02, *Paracoccus* sp. 4M6 or *Algibacter* sp. 1M6. The plates were incubated at 25°C for 48 h to allow the diffusion of SN_{D41} molecules from the disc into the agar medium and the growth of bacteria. The presence of a bactericidal compound would lead to clear halos of growth inhibition around the discs. Alternatively, *Vibrio* sp. D02, *Paracoccus* sp. 4M6 or *Algibacter* sp. 1M6 grown overnight in VNSS were inoculated into SN_{D41} at an OD_{600} of 0.25, incubated for 30 min or 2 h at 25°C or overnight at 4°C, and bacteria were centrifuged and resuspended into fresh VNSS medium. Dilutions were then plated onto VNSS agar and the number of colony forming units (CFU) was counted after growth for 48 h. Experiments were performed three times.

Statistical analyses

All tests in microtiter plates were carried out in six replicates and the results were averaged. The standard deviations (SDs) were calculated with a matricial program using Matlab software (Mathworks Inc., Natick, Massachusetts, USA). The statistical significance between different experimental conditions was determined by the paired Student's *t*-test (*p* values of <0.05 were considered significant).

Results

Mixed *Pseudoalteromonas* biofilms: inhibition of strain 3J6 by strain D41

Pseudoalteromonas sp. 3J6 was previously shown to secrete anti-biofilm molecules which were active against all tested marine bacteria except other *Pseudoalteromonas* isolates such as the *Pseudoalteromonas* sp. strain D41 (Dheilly et al. 2010). This led to the hypothesis that *Pseudoalteromonas* sp. 3J6 allows the growth of *Pseudoalteromonas* in marine biofilms

while impairing the biofilm development of other marine bacteria. In order to examine the behavior of two *Pseudoalteromonas* isolates within a single biofilm, advantage was taken of the GFP tagging of *Pseudoalteromonas* sp. 3J6 with plasmid pCJS10 (Dheilly et al. 2010). This allowed coinoculation of *Pseudoalteromonas* sp. 3J6(pCJS10) and D41 in a flow-cell chamber, and the two strains within the resulting biofilm could be distinguished by staining all bacteria with Syto 61 red and recording the red (all bacterial cells) and green (3J6[pCJS10] strain only) fluorescence. In an overlay of the two images, *Pseudoalteromonas* sp. 3J6(pCJS10) appeared yellow or green, whereas *Pseudoalteromonas* sp. D41 was red. The biofilm obtained after growth for 48 h was homogeneous (Figure 1A) with an average thickness of 2.7 μm (Table 2). It contained very few scattered green fluorescent bacteria (Figure 1A), which accounted for only 6.9% of the total biovolume (Table 2). Since it had been previously verified that pCJS10 was stably maintained in *Pseudoalteromonas* sp. 3J6 without selective pressure in the biofilm culture conditions used (Dheilly et al. 2010), this indicated that the *Pseudoalteromonas* sp. D41 strain was largely predominant in a biofilm resulting from the coinoculation of the two strains. Consistently, this biofilm was more similar to a single-strain biofilm of strain D41 (Figure 1B) than of strain 3J6 (Figure 1C). These results therefore showed that *Pseudoalteromonas* sp. D41 is able to out-compete *Pseudoalteromonas* sp. 3J6 during biofilm growth. This could not be simply

explained by a faster growth of D41 strain, since both strains exhibited similar doubling times (45 and 53 min for D41 and 3J6, respectively) in liquid VNSS cultures at 20°C (Dheilly et al. 2010; data not shown).

Inhibition of biofilm formation by Pseudoalteromonas sp. D41 exoproducts

Whether the supernatant of a liquid *Pseudoalteromonas* sp. D41 culture (SN_{D41}) could prevent the formation of biofilm by *Pseudoalteromonas* sp. 3J6 was then examined. This strain was resuspended either in SN_{D41} or in artificial marine water as a positive control. The bacterial suspensions were introduced into flow-cell chambers and incubated for 2 h to allow bacterial attachment onto the glass surface. Flows of VNSS medium free of bacterial exoproducts were then applied for 48 h to allow biofilm formation, and biofilms were stained with Syto 61 red and 0.5 μM Sytox green (live-dead staining) to evaluate the cell viability. The presence of SN_{D41} during the adhesion step partially prevented development of the *Pseudoalteromonas* sp. 3J6 biofilm (Figure 2, compare panels A and B). The average thickness and the biovolume were 7.8 and 9.3 fold lower, respectively, than in the positive control without SN_{D41} (Table 3). The live-dead staining showed that the proportion of nonviable cells was very low in both *Pseudoalteromonas* sp. 3J6 biofilms (Figure 2A and B, green and yellow bacteria in the upper top and lower images, respectively),

Table 1. Strains used in this study.

Strain	Origin	Reference and/or source
<i>Pseudoalteromonas</i> sp. 3J6	Morbihan gulf, France ^a	Grasland et al. (2003)
<i>Paracoccus</i> sp. 4M6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Bacillus</i> sp. 4J6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Vibrio</i> sp. D01	Bay of Brest, France ^b	Pradier et al. (2005)
<i>Algibacter</i> sp. 1M6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Colwellia</i> sp. 4J3	Morbihan gulf ^a	Grasland et al. (2003)
<i>Sulfitobacter</i> sp. 8J6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Vibrio</i> sp. D66	Bay of Brest ^b	IFREMER collection
<i>Alteromonas</i> sp. 1J3	Morbihan gulf ^a	Grasland et al. (2003)
<i>Pseudoalteromonas</i> sp. 3J3	Morbihan gulf ^a	Grasland et al. (2003)
<i>Pseudoalteromonas</i> sp. D41	Bay of Brest ^c	Pradier et al. (2005); Leroy et al. (2008a, 2008b)
<i>Micrococcus</i> sp. 5J6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Paracoccus</i> sp. 3M6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Pseudoalteromonas</i> sp. 5M6	Morbihan gulf ^a	Grasland et al. (2003)
<i>Vibrio</i> sp. D02	Bay of Brest ^b	IFREMER collection
<i>Vibrio</i> sp. D22	Bay of Brest ^b	IFREMER collection
<i>Alteromonas</i> sp. 2J3	Morbihan gulf ^a	Grasland et al. (2003)
<i>Pseudoalteromonas atlantica</i> T6c (mucoïd form)		Yaphe (1957)
<i>Pseudoalteromonas haloplanktis</i> TAC125	Antarctic ^e	Médigue et al. (2005)
<i>Pseudoalteromonas tunicata</i> D2	Gullmarsfjorden ^d	Holmström et al. (1998), DSMZ14096

Note: ^aIsolated from a glass slide immersed for 3 or 6 h in the Morbihan gulf, France; ^bIsolated from a glass slide immersed for 6 h in the Bay of Brest, France; ^cIsolated from a Teflon coupon immersed for 24 h in the Bay of Brest, France; ^dIsolated from an adult tunicate *Ciona intestinalis* collected from a depth of 10 m in Gullmarsfjorden, Sweden; ^eIsolated from an Antarctic coastal sea water sample collected near the French Antarctic station Dumont d'Urville, Terre Adelie.

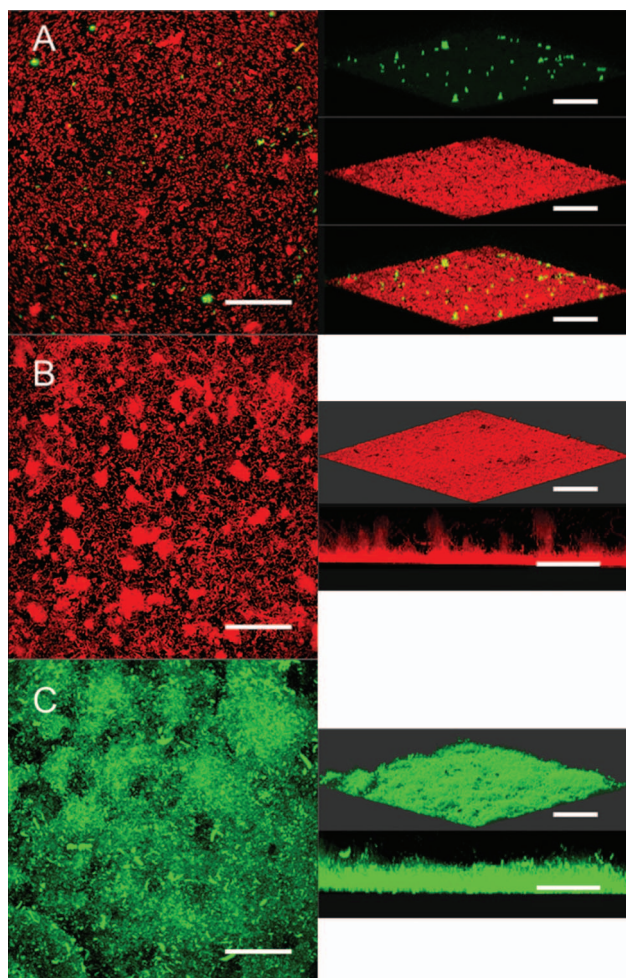


Figure 1. CLSM images of *Pseudoalteromonas* sp. 3J6(pCJS10) and D41 biofilms grown on glass surfaces at 20°C for 48 h under a flow of VNSS medium. (A) Biofilm resulting from the co-inoculation of the two strains. Maximal projections in the x - y plane (left) and three-dimensional (3D) shadow projections (right) are displayed. Green (*Pseudoalteromonas* sp. 3J6[pCJS10]) and red (all cells) fluorescence are shown separately (right, top and middle, respectively) or overlaid (left and right, bottom). (B) and (C) Single-strain biofilms of *Pseudoalteromonas* sp. D41 and of *Pseudoalteromonas* sp. 3J6(pCJS10), respectively. Maximal projections in the x - y plane (left), three-dimensional (3D) shadow projections (right, top) and projections in the x - z plane (right, bottom) are presented. The images are representative of at least two independent experiments from which a total of 12 to 18 image stacks were obtained. Bars = 50 μ m.

showing that SN_{D41} did not affect cell viability in the biofilm.

To examine whether SN_{D41} was active specifically against *Pseudoalteromonas* sp. 3J6 or if its action could be broader, the same experiments were performed on another marine bacterium, *Paracoccus* sp. 4M6. This strain was chosen because it forms thick biofilms

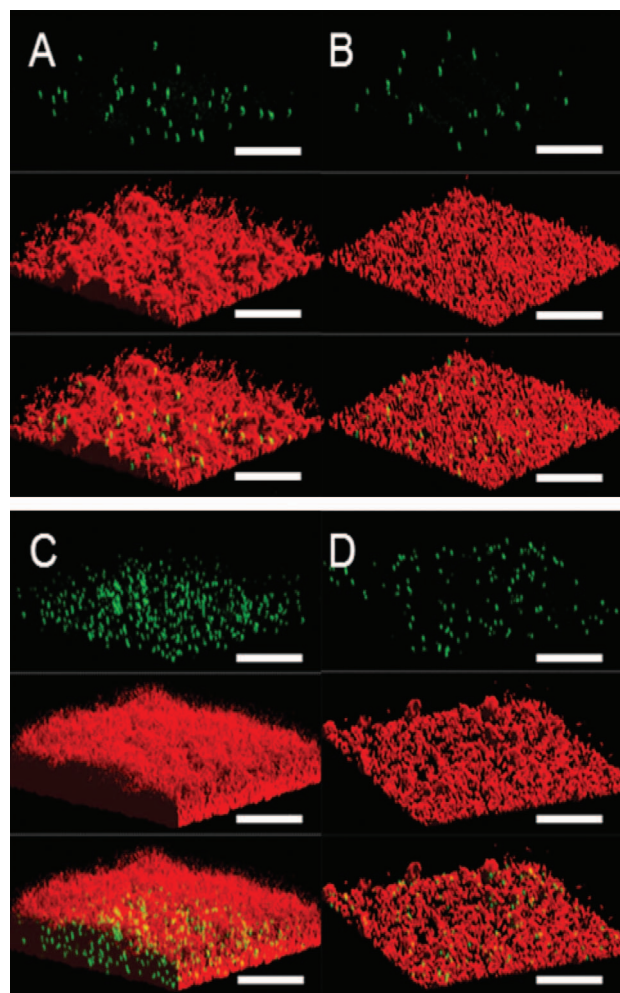


Figure 2. Effect of *Pseudoalteromonas* sp. D41 culture supernatant (SN_{D41}) on biofilm formation by *Pseudoalteromonas* sp. 3J6 (A and B) and *Paracoccus* sp. 4M6 (C and D). Biofilms were grown in flow-cell chambers for 48 h after a 2-h adhesion step without SN_{D41} (A and C) or in the presence of SN_{D41} (B and D). Biofilms were stained with Syto 61 red (staining of all bacteria) and Sytox green (staining of nonviable cells only). 3D shadow projections are shown for each biofilm (green fluorescence at the top, red fluorescence in the middle, and green and red overlay at the bottom of each panel). Bars = 62 μ m.

(Dheilly et al. 2010; Figure 2C and Table 3) and is highly sensitive to the anti-biofilm activity of strain 3J6 (Dheilly et al. 2010). *Paracoccus* sp. 4M6 biofilms displayed 4.3 and 3.6 fold lower average thicknesses and biovolumes, respectively, upon exposure to SN_{D41}, whereas the cell viability was not affected (Figure 2C and D; Table 3). SN_{D41} thus displayed anti-biofilm activity against both *Pseudoalteromonas* sp. 3J6 and *Paracoccus* sp. 4M6, showing that SN_{D41} targets various bacteria. The anti-biofilm activity was also produced by *Pseudoalteromonas* sp. D41 grown as a single-strain biofilm on the inner hydrophobic surface

of silicone tubing under a continuous flow of VNSS medium: the medium exiting from the tubing displayed a similar level of anti-biofilm activity as SN_{D41} from a liquid culture (data not shown). All the subsequent experiments were performed with anti-biofilm activities obtained from liquid culture supernatants since it was the easiest way to obtain the required quantity of anti-biofilm material.

Set up and validation of a microtiter plate assay of the anti-biofilm activity

The spectrum of action of *Pseudoalteromonas* sp. 3J6 culture supernatant (SN_{3J6}) was previously determined by observing its effect on the development of single-strain biofilms in flow-cell chambers (Dheilly et al. 2010). Although this method is highly informative as can be seen above, it is labor intensive and a quicker method in microtiter plates was developed. The latter method is detailed in the 'Materials and methods' section. Briefly, bacteria suspended in a culture supernatant (SN_{3J6} or SN_{D41}) were incubated for 2 h in a microtiter plate well (adhesion step), the bacterial suspension was replaced with fresh VNSS medium,

and the plate was incubated for 24 h to allow attached cells to develop biofilms (growth step). These two steps were performed at 25°C without shaking. Bacteria in biofilms were then quantified after crystal violet staining (Pitts et al. 2003). The PI, taking into account blank wells (no bacterium) and positive controls (artificial marine water replacing the anti-biofilm supernatant), was calculated as described in 'Materials and methods'.

Since the anti-biofilm activity of SN_{3J6} had been previously examined against nine marine bacterial strains in flow cells (Dheilly et al. 2010), the microtiter plate assay was first performed with SN_{3J6} against the same nine strains in order to compare the two methods. They yielded consistent data for seven strains: both methods showed that SN_{3J6} impaired biofilm development of the *Algibacter*, *Paracoccus*, *Cowellia*, *Sulfitobacter*, and *Vibrio* sp. D66 isolates, with similar inhibition levels, and that SN_{3J6} had no effect (PI < 10) on biofilm formation by *Pseudoalteromonas* sp. 3J3 and D41 (Table 4). However, divergent results were obtained with two strains, viz. *Vibrio* sp. D01 and *Alteromonas* sp. 1J3. Whereas biofilm formation by *Vibrio* sp. D01 was strongly inhibited by SN_{3J6} in a flow-cell, it was barely affected in microtiter plates (Table 4). Without SN_{3J6}, this strain formed biofilms up to 45 µm thick with one of the highest biovolume (14 µm³ µm⁻²) on the hydrophilic glass slide of the flow cell chamber (Dheilly et al. 2010). By contrast, its ability to develop a biofilm on the hydrophobic polystyrene surface of the microtiter plate was very weak: the OD₅₉₀ values after crystal violet staining of *Vibrio* sp. D01 biofilms (control wells) did not significantly differ from the OD₅₉₀ of blank wells. The microtiter plate used was therefore not appropriate for growing *Vibrio* sp. D01 biofilms. This led to a definition of a threshold of 0.2 for the OD₅₉₀ value after crystal violet staining of control wells, above which the assay was significant. This value was >0.2 for all the other tested strains in Table 4. Regarding *Alteromonas* sp. 1J3, flow-cell assays revealed that SN_{3J6} affected this strain in a peculiar way: it did not impair its ability to form biofilms, but it led to a much higher proportion of nonviable cells

Table 2. COMSTAT analyses of single- and two-strain biofilms after growth for 48 h in flow-cell chambers.

Biofilm	Maximal thickness (µm) ^a	Average thickness (µm) ^a	Biovolume (µm ³ µm ⁻²) ^a
Single-strain			
<i>Pseudoalteromonas</i> sp. 3J6(pCJS10)	21	4.8	2.6
<i>Pseudoalteromonas</i> sp. D41	29	3.6	2
Two-strains			
<i>Pseudoalteromonas</i> sp. 3J6(pCJS10) + D41	12	2.7	2.9
<i>Pseudoalteromonas</i> sp. 3J6(pCJS10)	12	0.5	0.2
<i>Pseudoalteromonas</i> sp. D41	7	2.7	2.7

Note: ^aAverages of data from three independent experiments, with SDs < 10% of each value.

Table 3. COMSTAT analyses of single-strain biofilms grown in flow-cell chambers for 48 h after a 2 h adhesion step or in the presence and absence of SN_{D41}.

Biofilm	Maximal thickness (µm) ^a		Average thickness (µm) ^a		Biovolume (µm ³ µm ⁻²) ^a	
	–SN _{D41}	+SN _{D41}	–SN _{D41}	+SN _{D41}	–SN _{D41}	+SN _{D41}
<i>Pseudoalteromonas</i> sp. 3J6	36.5	12.7	34.4	4.4	14.9	1.6
<i>Paracoccus</i> sp. 4M6	70.7	21.4	66.4	15.3	35.3	9.7

Note: ^aAverages of data from two independent experiments, with SDs < 10% of each value.

within the biofilms (80% vs 8.5% without SN_{3J6}) (Dheilly et al. 2010). With the microtiter plate assay, the development of *Alteromonas* sp. 1J3 biofilms was strongly inhibited (Table 4). Thus, both methods showed that SN_{3J6} acted against *Alteromonas* sp. 1J3, albeit at different levels.

Overall, the two methods were in excellent agreement for 7 strains, partially convergent for *Alteromonas* sp. 1J3, and the microtiter plates used were not appropriate for *Vibrio* sp. D01. This showed that the microtiter plate assay could be used with

confidence, providing that the strains are able to develop a biofilm onto polystyrene.

SN_{D41} spectrum of action

The microtiter plate assay was used to determine the SN_{D41} spectrum of action. The SN_{D41} activity was assayed against 18 strains consisting of 15 strains which were previously isolated from the coasts of Brittany (France), and three well characterized *Pseudoalteromonas* strains of various origins, viz. *P. atlantica* T6c (mucoïd form), *P. haloplanktis* TAC125, and *P. tunicata* D2 (Table 1). All these strains were able to form biofilms on the polystyrene of microtiter plates, which led to OD₅₉₀ values >0.2 after crystal violet staining. Table 5 shows the percentages of biofilm inhibition, on the basis of which five groups of bacteria could be distinguished.

Group 1 contains the most sensitive bacteria, with a PI ranging from 68 to 90. This group contains six strains belonging to various genera, viz. *Paracoccus* sp. 4M6, *Vibrio* sp. D02, *Colwellia* sp. 4J3, *Algibacter* sp. 1M6, *Alteromonas* sp. 1J3 and *Micrococcus* sp. 5J6 (Table 5). Group 2 contains bacteria the ability of which to develop a biofilm was less strongly impaired by SN_{D41}, with PIs between 43 and 54. This group is composed of four *Pseudoalteromonas* strains, viz. the 3J6 isolate, and the *P. tunicata*, *P. atlantica*, and *P. haloplanktis* strains (Table 5). Strains of Group 3 were weakly inhibited, since the PIs were within the 20–28 range. These strains are *Bacillus* sp. 4J6,

Table 4. Comparison of flow-cell and microtiter plate assays of SN_{3J6} activity against marine bacteria.

Strain	Percentage inhibition	
	Flow cell ^a	Microtiter plate ^b
<i>Algibacter</i> sp. 1M6	99.8	74.6 ± 2.8
<i>Paracoccus</i> sp. 4M6	85.6	87.6 ± 1.3
<i>Vibrio</i> sp. D01	95.2	12.7 ± 4.9
<i>Colwellia</i> sp. 4J3	95.4	83.6 ± 1.2
<i>Sulfitobacter</i> sp. 8J6	67.0	40.6 ± 5.2
<i>Vibrio</i> sp. D66	87.5	53.1 ± 4.2
<i>Alteromonas</i> sp. 1J3	9.4	77.4 ± 1.8
<i>Pseudoalteromonas</i> sp. 3J3	5.1	0.9 ± 7.2
<i>Pseudoalteromonas</i> sp. D41	9.5	0.4 ± 1.8

Note: ^aPercentage inhibition was calculated from biovolume data reported in Dheilly et al. (2010) as follows: [(biovolume without SN_{3J6} – biovolume with SN_{3J6})/biovolume without SN_{3J6}] × 100. COMSTAT analyses were performed on three independent experiments, and SDs were < 10% of each biovolume value (Dheilly et al. 2010); ^bPercentages of inhibition and SDs were calculated as indicated in 'Materials and methods' from six replicates.

Table 5. SN_{D41} and SN_{3J6} spectra of action determined using the microtiter plate assay.

Group ^a	Strain	SN _{D41} (PI ± SD)	SN _{3J6} (PI ± SD) ^b
1. High inhibition level	<i>Paracoccus</i> sp. 4M6	89.4 ± 1.7	87.6 ± 1.3
	<i>Vibrio</i> sp. D02	84.3 ± 0.7	69.4 ± 0.3
	<i>Colwellia</i> sp. 4J3	80.9 ± 1.1	83.6 ± 1.2
	<i>Algibacter</i> sp. 1M6	80.3 ± 1.5	74.6 ± 2.8
	<i>Alteromonas</i> sp. 1J3	77.5 ± 1.8	77.4 ± 1.8
	<i>Micrococcus</i> sp. 5J6	68.7 ± 1.9	70.0 ± 2.1
2. Mild inhibition level	<i>Pseudoalteromonas</i> sp. 3J6	53.9 ± 3.3	0.6 ± 5.1
	<i>P. tunicata</i> D2	53.1 ± 9.3	47.3 ± 9.7
	<i>P. atlantica</i> T6c (mucoïd form)	47.7 ± 5.0	47.8 ± 6.1
	<i>P. haloplanktis</i> TAC125	43.5 ± 5.6	38.7 ± 5.8
3. Low inhibition level	<i>Bacillus</i> sp. 4J6	28.0 ± 7.3	30.1 ± 6.1
	<i>Alteromonas</i> sp. 2J3	27.6 ± 4.4	1.6 ± 6.0
	<i>Sulfitobacter</i> sp. 8J6	20.2 ± 6.1	40.6 ± 5.2
4. Non sensitive strains	<i>Vibrio</i> sp. D22	8.5 ± 8.8	75.5 ± 4.2
	<i>Pseudoalteromonas</i> sp. 3J3	5.5 ± 5.9	0.9 ± 7.2
	<i>Paracoccus</i> sp. 3M6	4.2 ± 6.0	–2.4 ± 4.7
	<i>Pseudoalteromonas</i> sp. 5M6	2.3 ± 1.5	0.8 ± 0.9
	<i>Pseudoalteromonas</i> sp. D41	0.2 ± 1.6	0.4 ± 1.8
5. Stimulated strain	<i>Vibrio</i> sp. D66	–21.4 ± 5.2	53.1 ± 4.2

Note: ^aThe groups were distinguished on the basis of the percentage inhibition (PI) of biofilm formation in the presence of SN_{D41}; ^bValues in bold highlight differences between the action spectra of SN_{3J6} and SN_{D41}.

Alteromonas sp. 2J3, and *Sulfitobacter* sp. 8J6 (Table 5). Group 4 includes strains which were considered to be unaffected in their ability to form biofilms, with PIs < 10. This group is composed of *Vibrio* sp. D22, *Paracoccus* sp. 3M6, two *Pseudoalteromonas* isolates from the Brittany coasts: strains 3J3 and 5M6, and *Pseudoalteromonas* sp. D41 itself (Table 5). Finally, group 5 is defined by only one strain, *Vibrio* sp. D66. Its negative PI indicates that SN_{D41} enhanced the capability of this strain to form a biofilm (Table 5). An increase of $21.4 \pm 5.2\%$ of bacteria in biofilms which developed in wells containing SN_{D41} was measured (Student test, $p \ll 0.05$).

Altogether, most of the tested bacteria were affected. SN_{D41} impaired biofilm formation by 13 strains (Groups 1, 2 and 3) out of 18. The sensitive strains belong to various taxonomic genera. Three genera were represented by several strains. Whereas the two *Alteromonas* isolates were sensitive to SN_{D41}, the three *Vibrio* strains each displayed a different behaviour in the presence of SN_{D41} since isolates D02, D22, and D66 were strongly inhibited, unaffected, and stimulated, respectively (Table 5). The *Pseudoalteromonas* strains were either mildly inhibited (four strains defining Group 2) or unaffected (three strains including D41 itself in Group 4).

Extension of SN_{3J6} spectrum of action and comparison with SN_{D41}

The microtiter plate assay was also used to broaden knowledge of the SN_{3J6} spectrum of action and to compare it to that of SN_{D41}. Table 5 shows that 13 strains out of 18 were sensitive to SN_{3J6}. Whereas the two *Pseudoalteromonas* strains tested previously (3J3 and D41) were unaffected by SN_{3J6} (Dheilly et al. 2010), the new data revealed that *P. tunicata*, *P. atlantica*, and *P. haloplanktis* were all mildly inhibited by SN_{3J6}. SN_{D41} and SN_{3J6} had similar effects on most of the tested strains. Four strains displayed different behaviours in the presence of the two supernatants: *Vibrio* sp. D66 was stimulated by SN_{D41} and mildly inhibited by SN_{3J6}, *Vibrio* sp. D22 was unaffected by SN_{D41} and strongly inhibited by SN_{3J6}, *Alteromonas* sp. 2J3 was weakly inhibited by SN_{D41} and unaffected by SN_{3J6}, and *Pseudoalteromonas* sp. 3J6 was mildly inhibited by SN_{D41} whereas it was unaffected by its own culture supernatant (Table 5).

Physico-chemical characteristics of SN_{D41} and SN_{3J6} anti-biofilm compounds

The above differences in their action spectra suggested that the anti-biofilm molecules present in SN_{D41} and

SN_{3J6} are not identical. To further examine this hypothesis, whether these molecules retained their antibiofilm activities after submitting SN_{D41} and SN_{3J6} to various treatments was tested. The SN_{D41} and SN_{3J6} activities were both heat- and proteinase K-sensitive (Table 6), indicating that the anti-biofilm molecules are likely to be proteinaceous. Since DMSO could be used in the course of future purification procedures, adding various DMSO concentrations was tested to determine whether this could affect their activity. Table 6 shows that 0.1% of DMSO reduced about 2-fold the SN_{3J6} activity, which was completely lost at a concentration of 1%. In contrast, SN_{D41} activity was fully retained in 1% DMSO, and was only partially impaired by 10% DMSO. Finally, it was previously reported that the anti-biofilm molecules from SN_{3J6} could be retained on a Sep-Pak Plus C₁₈ cartridge (Waters Corporation, Milford, MA) and eluted with 50% acetonitrile (Dheilly et al. 2010). In contrast, these authors were unable to elute active SN_{D41} antibiofilm molecules from a Sep-Pak Plus C₁₈ cartridge, either with acetonitrile or with other solvents (methanol or butanol). These different behaviours strengthen the hypothesis that the anti-biofilm molecules of SN_{D41} and SN_{3J6} are not identical.

Assay of SN_{D41} antibacterial activity

SN_{3J6} has been shown to be devoid of antibacterial activity against free-living bacteria (Dheilly et al. 2010). This prompted investigation of whether it is the same for SN_{D41}. *Algibacter* sp. 1M6, *Paracoccus* sp. 4M6 and *Vibrio* sp. D02 were selected as target organisms since all three are highly sensitive to the anti-biofilm activity of SN_{D41} (Table 5). Using the disc diffusion assay, neither crude SN_{D41} nor 100-fold concentrated SN_{D41} prevented the growth of the tested bacteria (data not shown). Furthermore, liquid culture

Table 6. Effect of various treatments on the anti-biofilm activities of SN_{D41} and SN_{3J6}.

Treatment	Percentage inhibition ^a	
	SN _{D41}	SN _{3J6}
None	83.2 $\pm$ 0.7	68.0 $\pm$ 1.0
50°C, 1 min	0	0
Proteinase K, 2 h	15.0 $\pm$ 4.0	0
0.1% DMSO ^b	82.6 $\pm$ 1.0	37.0 $\pm$ 1.0
1% DMSO ^b	79.3 $\pm$ 1.5	0
10% DMSO ^b	36.3 $\pm$ 3.0	0

Note: ^aPercentages of inhibition of SN_{D41} and SN_{3J6} were determined against the *Vibrio* sp. D02 and *Algibacter* sp. 1M6 strains, respectively, using the microtiter plate assay. The values and their SDs were calculated from six replicates as described in 'Materials and methods'; ^bdimethyl sulfoxide (DMSO) was added in the supernatants at the indicated final concentration.

assays were used to monitor the effect of SN_{D41} on the viability of planktonic bacteria. Incubation up to 2 h at 25°C or overnight at 4°C in SN_{D41} failed to reduce the number of cultivable bacteria (data not shown).

Discussion

It has previously been reported that the culture supernatant of the *Pseudoalteromonas* sp. strain 3J6 (SN_{3J6}) displays anti-biofilm activity (Dheilly et al. 2010). In the present study, it was found that the *Pseudoalteromonas* sp. isolate D41 also secretes such activity. The D41 activity was characterized, a microtiter plate assay was set up and validated by comparing with the previous CLSM observations of the SN_{3J6} effects, and the spectrum of action of SN_{D41} was determined and compared with that of SN_{3J6}. The originality highlighted regarding SN_{3J6} is its lack of antibacterial activity against free-living bacteria, making its activity specifically directed against biofilms (Dheilly et al. 2010). Similarly, SN_{D41} failed to display antibacterial activity. By contrast, most of the known anti-biofilm molecules are also antibacterial (bactericidal or bacteriostatic). This is for example the case for the antibacterial protein AlpP secreted by *P. tunicata* D2 (James et al. 1996; Rao et al. 2005, 2006). The *P. tunicata* D2 was used as a control, and both its anti-biofilm and its antibacterial activities were detected with the assays used (data not shown). Since antibacterial activities are more often searched for than anti-biofilm activities, it is possible that the latter remain under-estimated. Consistently, recent screenings of marine bacteria for anti-biofilm activities were successful (Nithya and Pandian 2010). SN_{3J6} and SN_{D41} are furthermore alike in their wide action spectra against Gram-negative bacteria, with 72% of sensitive strains for each supernatant (Table 5). The two spectra are, however, not identical, the most striking differences being the sensitivity of *Pseudoalteromonas* sp. 3J6 to SN_{D41} but not to its own culture supernatant, the sensitivity of *Vibrio* sp. D22 to SN_{3J6} but not to SN_{D41}, and the opposite effects for SN_{3J6} and SN_{D41} on *Vibrio* sp. D66 (Table 5). SN_{3J6} and SN_{D41} were equally efficient at impairing biofilm formation by *Paracoccus* sp. 4M6 (Table 5), but CLSM observations showed that SN_{3J6} led to a residual biofilm containing 100% of nonviable cells (vs 1.2% in the absence of SN_{3J6}) (Dheilly et al. 2010) whereas the viability was unaffected in the residual biofilms obtained with SN_{D41} (Figure 2C and D). These differences suggest that the anti-biofilm molecules secreted by *Pseudoalteromonas* sp. 3J6 and D41 might be related but are unlikely to be identical. This is supported by the observation that the anti-biofilm molecules from strains 3J6 and D41 displayed different behaviours in the presence of

DMSO and when attempts were made to elute them from Sep-Pak Plus C₁₈ cartridges. Very few natural molecules were reported to display anti-biofilm activity vs Gram-negative bacteria without being also antibacterial. Such molecules could be polysaccharides (Valle et al. 2006), biosurfactants (Das et al. 2009), or quorum sensing inhibitors (Rasmussen and Givskov 2006; Estrela et al. 2009). As discussed by Dheilly et al. (2010), none of these hypotheses was really consistent with the SN_{3J6} data. Their sensitivities to heat (50°C) and proteinase K indicate that the anti-biofilm molecules of both SN_{D41} and SN_{3J6} are likely to be proteinaceous. Understanding the mode of action of SN_{3J6} and SN_{D41} will require identification of the active molecules. Their purification is therefore under progress, using the microtiter plate assay to detect and quantify the anti-biofilm activities in the chromatography fractions.

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